

Skill lab

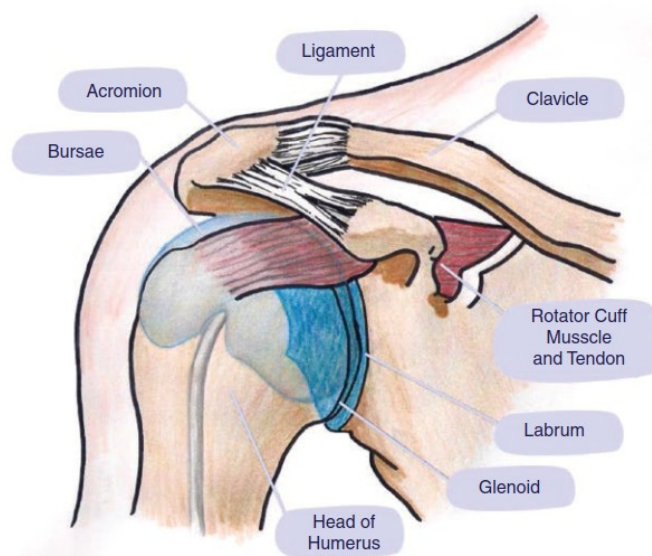
Unit 3: musculoskeletal system

Week 2: shoulder joint

Objectives:

- 1- describe the surface anatomy of the shoulder joint
- 2- explain the function and innervation of rotator cuff muscles
- 3- how to take a focused history from patient with shoulder pain
- 4- how to perform physical examination of the shoulder joint
- 5- how to interpret findings from history and physical examination from the patient (clinical reasoning)
- 6- how to differentiate between common nontraumatic causes of shoulder pain and manage them using illness script

The shoulder joint consists of the glenohumeral joint, acromioclavicular joint and subacromial space. Movement also occurs between the scapula and the chest wall. **The rotator cuff is composed of the supraspinatus, subscapularis, teres minor and infraspinatus muscles.** They and their tendinous insertions assist stability and movement, particularly abduction at the glenohumeral joint.



Pathology

Rotator cuff impingement syndrome is a painful disorder which is thought to arise from repetitive compression or rubbing of the tendons (mainly supraspinatus) under the coracoacromial arch. Normally, when the arm is abducted, the conjoint tendon slides under the coracoacromial arch.

If the arm is held persistently in abduction and then moved to and fro in internal and external rotation (as in cleaning a window, painting a wall or polishing a flat surface) the rotator cuff may be compressed and irritated as it comes in contact with the anterior edge of the acromion process and the coracoacromial ligament. This attitude (abduction, slight flexion and internal rotation) has been called the 'impingement position'.

The physical examination

Ask the patient to sit or stand and expose their shoulder completely. In addition to what was mentioned in general principles:

Look

- Size and position of the scapula: assess whether it is elevated, depressed or 'winged'.



Fig. 13.28 Right anterior glenohumeral dislocation. Loss of the normal shoulder contour.



Fig. 13.29 'Winging' of the left scapula. This caused by paralysis of the nerve to serratus anterior.

Feel

- sternoclavicular joint along the clavicle to the acromioclavicular joint, acromion and coracoid (2 cm inferior and medial to the clavicle tip) processes, the scapula spine and the biceps tendon in the bicipital groove. Extend the shoulder to bring the supraspinatus anterior to the acromion process, allowing palpation of the supraspinatus tendon.

Move:

Flexion, extension, internal and external rotation, abduction and adduction.

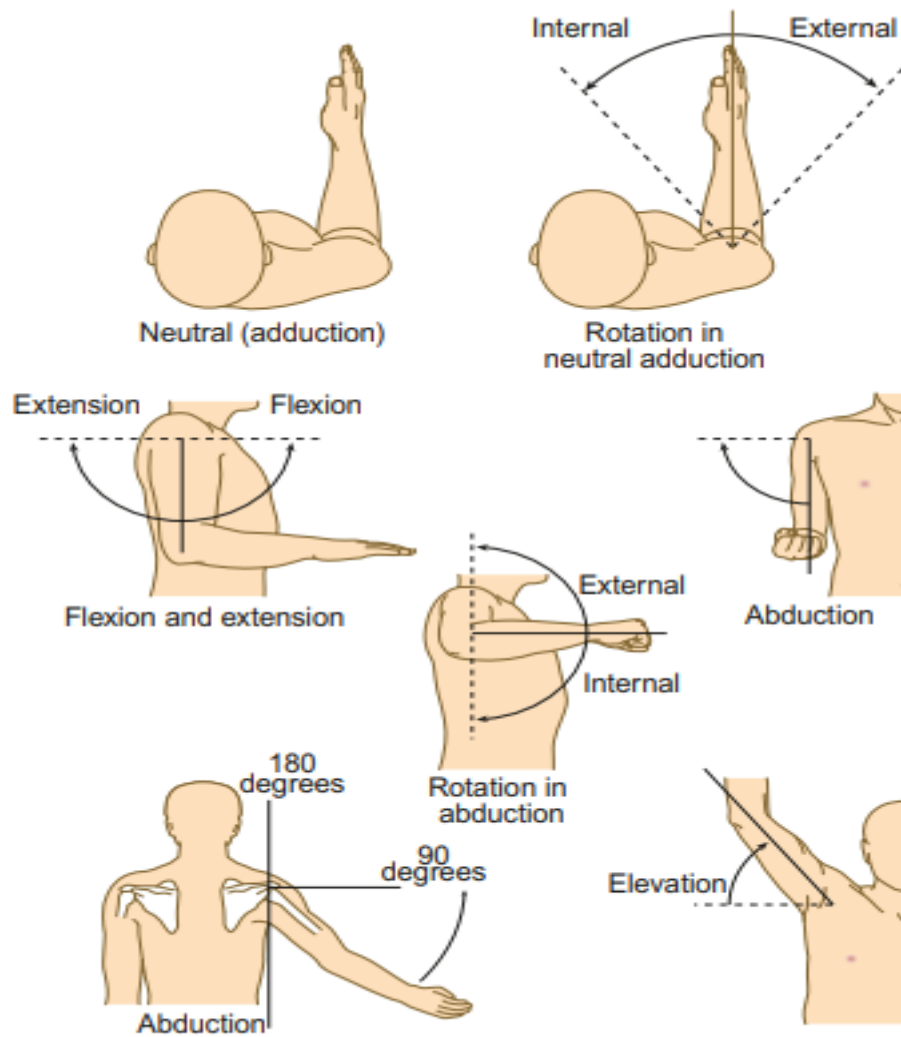


Fig. 13.30 Movements of the shoulder.

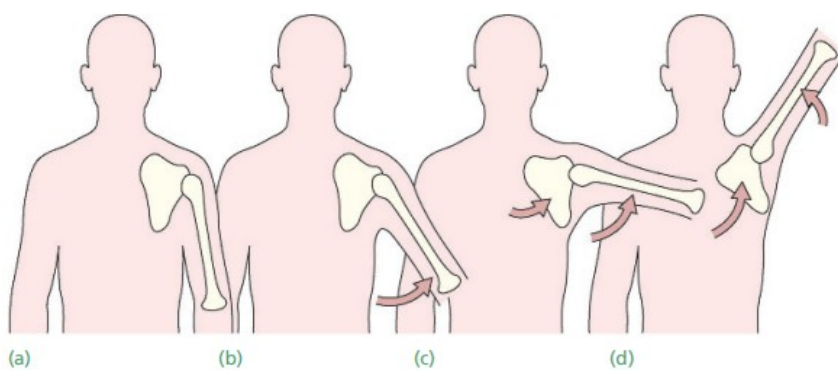


Figure 13.2 Scapulohumeral rhythm (a–c) During the early phase of abduction, most of the movement takes place at the glenohumeral joint. As the arm rises, the scapula begins to rotate on the thorax (c). In the last phase of abduction, movement is almost entirely scapulothoracic (d).

Special tests for impingement

Impingement (painful arc): Passively and fully abduct the patient's arm. Ask them to lower (adduct) it slowly. Pain occurring between 60 and 120 degrees of abduction occurs in painful arc. Passively abduct the internally rotated arm to 30–45 degrees. Ask them to continue to abduct their arm. Pain on active movement, especially against resistance, suggests impingement.

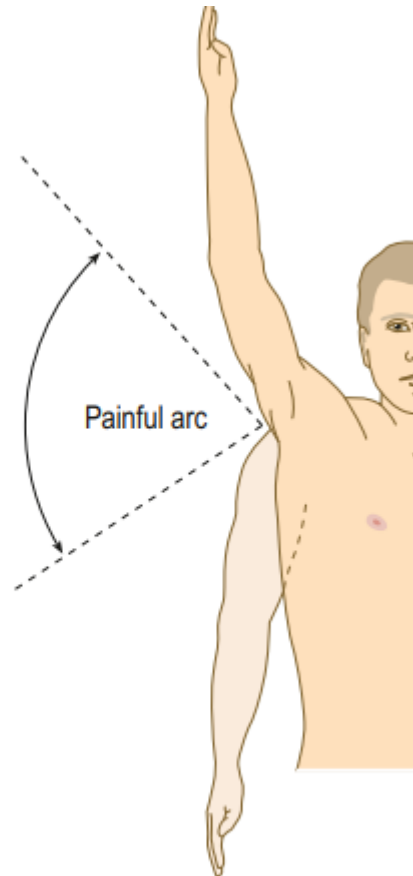


Fig. 13.27 Painful arc.

Neer test: The patient sits in a relaxed position with their elbow fully extended. Scapular rotation is prevented with one hand while the other abducts the arm in internal rotation. This causes the greater tuberosity to impinge against the acromion. A positive test is indicated by pain. In Neer's test, passive shoulder flexion from 90° to 180° exacerbates underlying narrowing of the passage made up by the acromion, coracoacromial ligament and humeral head, resulting in compression of its contents (the supraspinatus and infraspinatus tendons). Neer's impingement test is somewhat useful to exclude rotator cuff tendon impingement with a negative test.

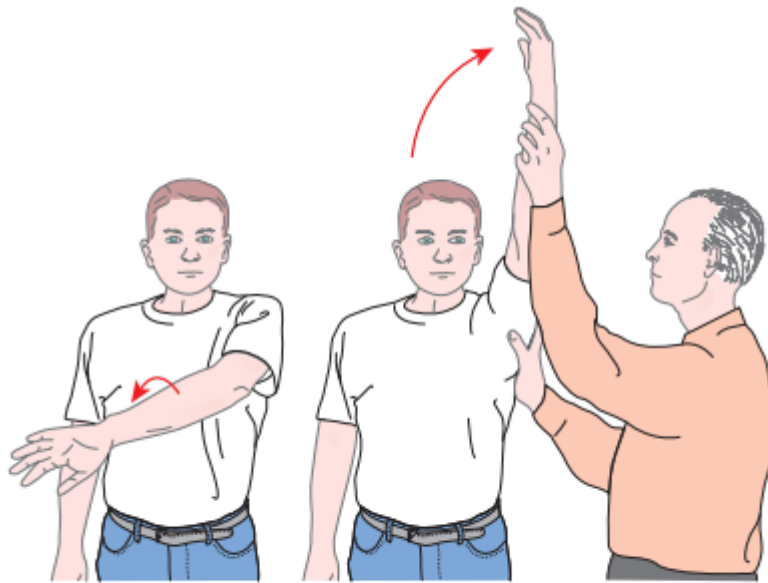


FIGURE 1.27
Neer's impingement test

Hawkins–Kennedy test: The patient is examined while sitting or standing, with their shoulder flexed at 90 degrees and their elbow flexed at 90 degrees, supported by the examiner to ensure maximal relaxation. The examiner forcefully rotates the arm internally. Reproduction of the patient's pain is a positive sign. Hawkins' impingement test exacerbates narrowing in the coracoacromial space and will worsen pre-existing impingement of the tendons and muscles when present. This manoeuvre will also elicit tenderness when rotator cuff tendonitis is present, due to mechanical forces or compression on the injured tendon or muscle.

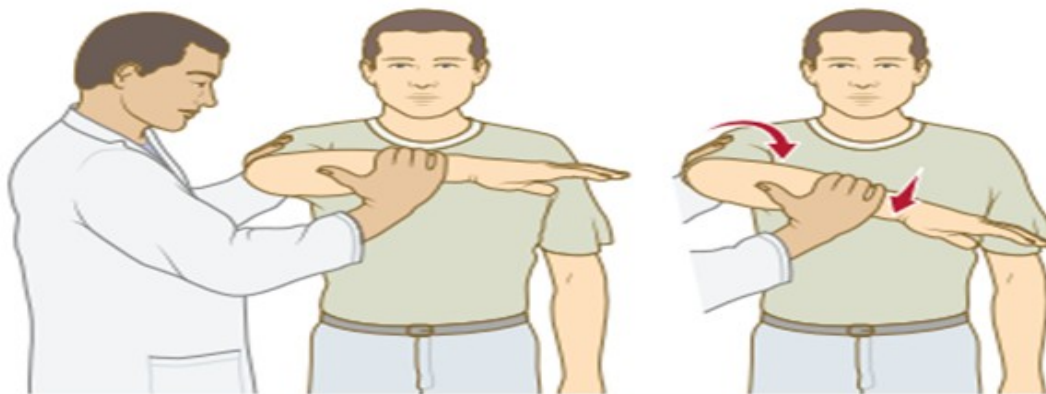


Fig. 13.32 Hawkins–Kennedy test for shoulder impingement.

Checklist for shoulder joint examination

		Not done	Done
Introduction and rapport	- Greet & Introduce self to the patient - Washes hands before and after examination - Determines how the patient wishes to be addressed - Explains nature of the examination to the patient - Obtains consent and Ensures privacy	0	1
Look	- Inspects all around the joint for discoloration, scars, any bony or joint swellings or deformities - Compares with contra lateral limb/joint (asymmetry) - Inspects the joint Position From the front, sides: Subacromion, back and above	0	1
Feel	- Palpates the joint surface landmarks - Palpates the supraspinatus, infraspinatus, deltoid and biceps areas for tenderness. * Comments on the Temperature, Swelling, Tenderness	0	2
Move	- Tests active ROM: Forward Flexion, Extension, Abduction, Adduction, Internal Rotation, External Rotation and Circumduction.	0	2
	- Tests passive ROM	0	1
Special tests	Painful arc, Neer test	1	2
Closure	- Examines Joint above and joint below - Check distal pulse - Neurological assessment	0	1
	Thank the patient		
Total mark		/10	

Shoulder pain

Shoulder pain, which is one of the most common musculoskeletal maladies, may arise from diverse causes. One of the most complex and mobile joints of the body, the shoulder is traversed by muscle, tendon, and bone and is surrounded by major neurovascular structures, all of which may serve as potential sources of local and referred pain.

13.15 Causes of shoulder girdle pain	
Rotator cuff	
• Degeneration • Tendon rupture	• Tendonitis • Calcification
Subacromial bursa	
• Inflammation due to inflammatory arthritis, injury or overuse	• Calcification
Capsule	
• Inflammation (Adhesive capsulitis, polymyalgia rheumatica)	
Head of humerus	
• Tumour • Osteonecrosis	• Fracture/dislocation
Joints	
• Glenohumeral, sternoclavicular: • Inflammatory arthritis, osteoarthritis, dislocation, infection	• Acromioclavicular: • Subluxation, osteoarthritis

Case scenario: 36-year-old male, presents with shoulder pain for 4 weeks duration.

- 1- take a focused history from him
- 2- perform relevant physical examination
- 3- formulate a differential diagnosis

Case:

A 42-year-old man with diabetes presents with progressive shoulder pain over the last 4 weeks. His pain increased on lying on the affected side and with lifting his arm above his head.

Joint examination reveals no swelling, but tenderness and painful limitation of shoulder motion. Normal cervical and elbow joint examination. Normal radial pulse. Normal muscle tone and bulk.

Illness Scripts:	<i>Differential #1</i>	<i>Differential #2</i>	<i>Differential #3</i>
Epidemiology			
Time Course			
Clinical Presentation			
Pathophysiology			

Labs/Imaging:

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Final Problem Representation/Diagnosis:

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SP instruction

Scenario	Ali hamza, 36 year old male
History of present illness	Chief complaint: right shoulder pain
Duration	4 weeks
Aggravating/alleviating factors	Overhead activity, lying on the affected side. Rest decreases pain
Character	Dull
Severity	6/10, moderate
Course	Increased over the last 2 weeks
Associated symptoms	NO stiffness, swelling, erythema. Warmth.... No fever, rash,,,,
Current life situation	(where does patient live/work, ...): house painting for 10 years
Personality	In distress
Past medical history	None . No trauma or surgery
Family history	None
Social history	Smoker. Not alcoholic
Medications	none. No drug allergy
Physical exam	Joint examination reveals no swelling, hotness, or redness. Tenderness at the anterior edge of acromion. Pain and limitation of active motion on abduction and external rotation with no weakness. Normal passive motion Positive painful arc, Neer and Hawkin tests Normal cervical and elbow joint examination. Normal radial pulse. Normal muscle tone and bulk